

Sathish Nallamekala

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PROFESSIONAL SUMMARY

Professional **Senior Data Engineer and Data Platform Engineer** with **6+** years of experience delivering end-to-end data engineering, cloud data platform, and analytics solutions. Proven ability to design, build, and optimize large-scale batch and real-time data pipelines supporting regulatory, analytical, and operational workloads.

I bring strong expertise in **PySpark, Databricks, Snowflake, Kafka, Airflow, and Azure**, with a focus on building scalable, secure, and high-performance data platforms.

As a Data Engineer, I have extensive experience modernizing legacy data systems, implementing data governance and quality frameworks, and supporting regulatory-grade reporting with high accuracy and reliability.

My background includes building real-time streaming solutions, optimizing Spark workloads, and designing dimensional data models to enable advanced analytics and reporting.

My career reflects a passion for cloud data engineering, automation, performance tuning, and delivering business-critical data solutions at enterprise scale.

TECHNICAL SKILLS

- **Programming & Querying:** SQL, Python, Scala
 - **Big Data & Processing:** PySpark, Apache Spark, Databricks, Delta Lake, Hadoop, Hive
 - **Streaming Technologies:** Kafka, Azure Event Hubs, AWS Kinesis
 - **ETL & Orchestration:** Apache Airflow, dbt, Talend, Informatica, AWS Glue
 - **Cloud Platforms:** Microsoft Azure (ADLS, Event Hubs), AWS (S3, EMR, Glue), Snowflake
 - **Data Warehousing & Modeling:** Star Schema, Snowflake Schema, Data Vault 2.0
 - **Data Governance & Quality:** Collibra, Great Expectations
 - **Security & Compliance:** Data Encryption, Masking, Tokenization, PCI, SOX
 - **DevOps & CI/CD:** Docker, Kubernetes, GitHub Actions
 - **Databases:** Snowflake, Oracle, Teradata
 - **Operating Systems:** Linux, Windows
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PROFESSIONAL EXPERIENCE

Senior Data Engineer | Morgan Stanley, USA | August 2024 – Present

Project: Enterprise Financial Data Platform & Regulatory Reporting

- Designed, built, and maintained scalable **PySpark ETL pipelines** in Databricks for large financial datasets including trading, exposure, and risk data.
- Developed **real-time data ingestion pipelines** using Kafka and Azure Event Hubs for market pricing and external vendor feeds.

- Designed and implemented **Snowflake data models** using Star and Snowflake schemas to support regulatory and analytical reporting.
- Automated end-to-end workflows using **Airflow and dbt**, ensuring accuracy, reliability, and audit readiness.
- Optimized Spark workloads using partitioning, broadcast joins, caching, and performance tuning techniques.
- Integrated complex financial data sources such as **Moody's and Murex** into downstream risk systems.
- Implemented **data security, encryption, masking, and metadata governance** using Collibra and cloud-native security controls.
- Supported large-scale migration from **Teradata and Oracle to Azure ADLS, Delta Lake, and Snowflake**.

Key Achievements:

- Reduced ETL processing time by **50%** through PySpark optimization.
- Achieved **99.97% data accuracy** for Basel III and CCAR reporting workflows.
- Reduced infrastructure costs by **35%** through cloud migration and optimization.

Technologies Used: Python, SQL, Scala, PySpark, Apache Spark, Databricks, Delta Lake, Snowflake, Azure, Azure Data Lake Storage (ADLS), Apache Kafka, Azure Event Hubs, Apache Airflow, dbt, Teradata, Oracle, Collibra, Data Masking & Encryption, Git, CI/CD, SOX & PCI Compliance Tools

Data Engineer | Capgemini, India | Jan 2019 – July 2023

Project: Retail & E-Commerce Big Data Platform

- Developed high-volume **PySpark ETL pipelines** processing structured and semi-structured retail datasets.
- Migrated legacy ETL workloads from **Informatica to Talend**, improving throughput and stability.
- Built **real-time streaming pipelines** using Kafka and AWS Kinesis for operational and analytical use cases.
- Designed cloud data architecture using **AWS S3 and Delta Lake** for scalable storage and analytics.
- Automated workflows using **Airflow and AWS Glue** to support BI and reporting systems.
- Designed dimensional data models using **Star and Snowflake schemas** for customer and product analytics.
- Optimized Hive queries through partitioning and bucketing strategies.
- Implemented **data quality validation** using Great Expectations.
- Deployed containerized Spark workloads using **Docker and Kubernetes**.

Key Achievements:

- Processed **1TB+ daily data volumes** across retail datasets.
- Improved system performance and reliability through automation and optimization.

- Enabled **85% ML forecasting accuracy** for retail demand prediction using MLlib.

Technologies Used : Python, SQL, PySpark, Apache Spark, Hadoop, Hive, Delta Lake, AWS, AWS S3, AWS EMR, AWS Glue, Apache Kafka, AWS Kinesis, Apache Airflow, Talend, Informatica, Great Expectations, Docker, Kubernetes, Git

EDUCATION

Master of Science in Computer Science (Auburn University at Montgomery, USA)

Bachelor of Technology in Electrical & Electronics Engineering(VVIT, India)

CERTIFICATIONS

- IBM Python 101 for Data Science
- Walmart Advanced Software Engineering – Forage
- SAP Technical Consulting – Forage
- Tata Data Visualization – Forage